

KHALED TUJJAR

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EDUCATION

University of Central Florida

Master of Science in Computer Science

Bachelor of Science in Computer Science | Minor: Mathematics

Expected August 2027

August 2024

EXPERIENCE

Sheefra Corporation | Orlando, FL

August 2023 – Present

Software Developer

- Co-built and maintain TransTrack, a full-stack PHP, MySQL, and JavaScript app saving staff 10+ hours weekly
- Cut report times from minutes to seconds by profiling, indexing, and restructuring its heaviest queries
- Designed versioned REST APIs with token-based authentication, serving three internal apps and 25+ daily users
- Co-led the production Linux server migration and administer its backups, monitoring, and security patching
- Developed a desktop time-clock app for remote workers using C++ and MySQL, reducing payroll errors by 80%

SKILLS

Languages: Python, C, C++, TypeScript, JavaScript, SQL, PHP, GLSL

Frameworks: React, React Native, Expo, FastAPI, PyTorch, OpenGL, TailwindCSS, pytest

Tools: Docker, Kubernetes, Kafka, Git, GitHub Actions, Linux, UNIX, bpfftrace, MySQL, CMake, npm, pip

PROJECTS

Anomaly Detection Platform | Python, PyTorch, Kafka, GitHub Actions github.com/KTujjar/anomaly-detection

- Designed and built a Kafka in-stream processing service flagging anomalies in live events via async FastAPI
- Prototyped an LSTM autoencoder against EWMA and 3-sigma baselines, quantifying parity limits on univariate data
- Containerized with Docker and deployed to Kubernetes with horizontal pod autoscaling
- Automated continuous integration and delivery in GitHub Actions, gating container publishes on 20 pytest cases

RocketDocs | TypeScript, React, TailwindCSS

github.com/ryanata/rocketdocs-frontend

- Shipped the single-page React application for an AI-powered documentation tool with a teammate
- Reviewed pull requests before merge and iterated with that teammate in a sprint-based agile workflow
- Created AI-assisted doc templates and a review queue, standardizing documentation style across 15 repositories

HID Behavior Detector | C++, Python, bpfftrace, CMake

github.com/KTujjar/hid-behavior-detector

- Built a Linux detector that flags suspicious command execution after an untrusted keyboard attaches
- Correlated bpfftrace exec, fork, and connect streams with HID attach events to tie activity to a device
- Scored runs in a C++17 analyzer with weighted heuristics and measured true and false positive rates

Pwdly | JavaScript, React Native, Expo

github.com/ryanata/password-manager

- Co-built a cross-platform password manager with vault encryption and master-password gating
- Unified mobile and web flows with shared components, eliminating duplicated UI code across platforms

N-Body Particle Renderer | C++, OpenGL, GLSL, CMake

github.com/KTujjar/nbody-renderer

- Implemented a real-time 3D gravitational simulation of 2,000 particles in C++17, OpenGL 3.3, and CMake
- Wrote the $O(n^2)$ physics core and render pipeline by hand, verifying accuracy with energy-conservation checks

LEADERSHIP & EXTRACURRICULARS

Knight Hacks — Won a hackathon award for prototyping an eye-tracking Snake game in 48 hours

Google Developer Student Clubs — Led a Git workshop and mentored 25 peers on version control

CERTIFICATIONS

Google Cybersecurity Certificate | Coursera

October 2024